

Case Report

The Value of Detecting and Monitoring ctDNA in Uveal Melanoma: Results of a Pilot Study and a Systematic Review

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Keywords

Circulating tumor DNA · Uveal melanoma · Variant allele frequency · Pilot study · Systematic review

Abstract

Introduction: This article presents the results of a prospective pilot study on circulating tumor DNA (ctDNA) monitoring in uveal melanoma (UM) patients, along with a systematic review of the literature to support and contextualize the findings. We investigated the clinical utility of ctDNA variant allele frequency (VAF) for early detection of tumor progression. **Case Presentations:** We performed serial ctDNA testing using the Ion AmpliSeq Cancer Hotspot Panel v2 and Ion Proton sequencing in 3 UM patients (two recurrent, one nonrecurrent) at four time points over 2 years. A systematic review of studies up to June 2024 was also conducted, identifying 32 additional UM patients. In recurrent patients, co-occurring mutations in *GNAQ*, *GNAS*, and *IDH1* were identified at the time of recurrence, with minimum VAFs of 0.51%, 0.25%, and 0.1%, respectively. In the nonrecurrent patient, pathogenic variants were absent, and VAFs of key genes declined over time. The systematic review supported the association of ctDNA alterations with recurrence and metastasis, identifying critical genes including *GNAQ/11*, *GNAS*, *IDH1*, *STK11*, *FGFR2*, and *EGFR*, many of which interact with the BAP1 signaling pathway. **Conclusion:** Serial ctDNA analysis may serve as a noninvasive tool for early detection of recurrence in UM. Co-occurrence of *GNAQ*, *GNAS*, and *IDH1* mutations may represent a molecular signature of recurrence. Further validation in larger cohorts is needed to define optimal gene panels and VAF thresholds for clinical use.

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Introduction

Uveal melanoma (UM) is one of the most common intraocular malignancies in adults, comprising 3%–5% of all melanomas. The predominant disease site originates from choroidal melanocytes [1, 2]. The incidence rate in North America and Europe is between 5.74 and 7.30 cases per 1 million [3].

More than 92% of UMs contain oncogenic mutations in the alpha subunit of the *GNAQ* and *GNA11* genes at positions Q209 and R183 [4–6], which induce cell proliferation, tumor growth, and progression by targeting the Akt/mTOR, Wnt/ β -catenin, yes-associated protein (YAP), and MAPK signaling pathways [7]. BAP1 (BRCA1 associated protein 1) is a tumor suppressor gene (TSG) located on 3p21 and is mutated in approximately 80% of UM metastasizing patients [8]. In addition to point mutations, UM is associated with chromosomal abnormalities and cytogenetic changes. Monosomy 3, loss of 1p, gain of 1q, gain of 6p, loss of 6q, loss of 8p, and gain of 8q are common chromosomal abnormalities in UM. These genetic changes can lead to an increase in genomic instability and increased metastasis [1, 9–11].

Currently, UM can be divided into four classes based on the Cancer Genome Atlas (TCGA) classification: class A (disomy 3 and 8q), class B (disomy 3/partial extra 8q), class C (monosomy 3/8q gain), and class D (monosomy 3/multiple 8q gain) [6, 12]. However, the American Joint Committee on Cancer (AJCC) staging considers clinical factors of the tumor, such as tumor size, extrascleral extension, and ciliary body involvement [13]. The constellation of information from these two classifications of anatomical features and genetic information improves the prognostication of patients with UM [14].

The local recurrence rate appeared to be 6% within 5 years of initial treatment [15]. Tumor size, specifically the largest basal diameter and thickness (TH), are two important prognostic factors that increase the likelihood of recurrence [16–21]. When the initial doses of radiation delivered via brachytherapy (BRT) at the time of diagnosis are insufficient to reach the base of the tumor, it can lead to frequent local and/or distant recurrences and an increased need for systemic therapy or enucleation [22]. Several studies suggest that after BRT for choroidal melanomas, there is often more than 90% local tumor control [23–25]. However, the rate of local tumor control after BRT can vary widely between ocular oncology centers, ranging from 0% to 22% [25–28].

Cell-free DNA (cfDNA) was introduced as a noninvasive biomarker in personalized tumor medicine [29]. Subsequently, circulating tumor DNA (ctDNA) was detected in the cfDNA components of a cancer patient's blood, which was caused by apoptosis and necrosis of tumor cells [30, 31]. ctDNA analysis enables the diagnosis and identification of noninvasive tumor detection, prediction of treatment response, monitoring of disease recurrence, and identification of mechanisms of resistance for the development of targeted therapies [32].

Today, there is increasing evidence that ctDNA has the ability to detect disease recurrence earlier or at the same time as conventional methods. The use of ctDNA remains limited to recurrence and monitoring of residual disease in clinically disease-free patients [33–36]. Therefore, it is unclear whether ctDNA can be detected at early stages and/or provide early evidence of disease advancement.

This manuscript presents both original data from a pilot study and a systematic review. The aim was to evaluate the clinical utility of ctDNA in detecting recurrence and metastasis in UM.

Case Presentation

In this prospective pilot study conducted between June 2020 and July 2022, ctDNA was measured in all 3 patients, with only two showing recurrence and one without clinical evidence of recurrence in patients with UM. Overall, we collected 12 blood samples from a cohort of three consecutive choroidal UM patients at four phases every 6 months following treatment, over a period of 24 months (baseline, 6th month, 12th month, and 24th month after treatment).

The variant allele frequency (VAF) was checked using the Ion Proton Sequencer in all phases. The ctDNA extraction samples of the patients were collected at each phase of the disease, and then the sequencing was done in the Genome Center of Nilou Laboratory using the Ion Torrent platform. Our study, with the ethics code IR.IUMS.REC.1398.730, was approved by the Iran University of Medical Sciences, and written informed consent was received from all participating cases in this study.

Indirect ophthalmoscopy, fundus photography, B-scan ultrasound evaluation, or ultrasonic biomicroscopy (Ellex, Adelaide, Australia) were analyzed for clinical status, along with abdominal magnetic resonance imaging (MRI) and chest computed tomography (CT) scans. The clinical features and demographics included age, gender, and clinical status, while tumor features included the largest basal diameter, height, location, laterality of disease, and primary treatment type (enucleation vs. plaque BRT).

Patient 1

A 57-year-old man was referred with posterior choroidal malignant melanoma of the left eye in July 2015. In indirect ophthalmoscopy of the fundus, a prominent pigmented mass was noted in the macula with dimensions of 14 × 13 mm by 8 mm in height, which was subjected to Ru-106 plaque radiotherapy. The patient also underwent a systemic examination for possible metastasis. Abdominal sonography and a triphasic CT scan of the liver revealed two hemangiomas with dimensions of 50 × 70 mm in the left lobe and another with a size of 7 × 16 mm. These lesions were confirmed by MRI. Because of inadequate response and suspicious recurrent disease, a secondary Ru-106 plaque was placed in July 2020, and the first sampling of the patient was done 2 days before placing the second plaque, and it was followed every 6 months until July 2022 (online suppl. Fig. S1; for all online suppl. material, see <https://doi.org/10.1159/000548442>).

Due to the presence of a new satellite lesion and the development of pigmented vitreous seeding (clumps) during follow-up, a vitreous biopsy was performed. The pathology report indicated the presence of active malignant pigmented cells. Therefore, after discussing with the patient, it was decided to schedule an enucleation. Following the enucleation, the tumor type was categorized as mixed spindle cell A and B based on the pathology findings. Blood tests, including lactate dehydrogenase (LDH), aspartate aminotransferase (AST), alanine transaminase (ALT), and alkaline phosphatase (ALP), were reported as normal. The tissue sample was subjected to multiplex ligation-dependent probe amplification (MLPA) for cytogenetic changes.

Patient 2

A 49-year-old man was referred for posterior choroidal malignant melanoma of the right eye, with a prominent pigmented mass on funduscopy. The tumor measured approximately 4 × 7 mm in basal diameter and 3.5 mm in height (online suppl. Fig. S2, S3). The first sampling was done in September of 2020, a few days before BRT, and serial sampling was done every 6 months until September 2022. A systemic examination was also conducted, which did not show any metastasis during these 2 years. One year after plaque radiotherapy, the UBM test revealed a slight increase in tumor TH. To further consolidate the

tumor, adjuvant transpupillary thermotherapy (TTT) was performed in phase 3 (12th month). In the last phase, the TH of the tumor was slightly reduced compared to phase 3. Additionally, blood tests (LDH, AST, ALT, and ALK) were reported as normal.

Patient 3

A 61-year-old woman was referred with a diagnosis of iris nevi in the right eye and a large vascularized mass in the left eye. The clinical exam and UBM test results revealed a large iridociliochoroidal malignant melanoma in the nasal part (online suppl. Fig. S4, S5). The patient refused fundus imaging in the early stages of treatment. Similar to other patients, the preliminary workup for metastasis was reported as normal. One week before BRT, the first sampling was done, and during the 24-month follow-up, UBM and clinical exams showed a progressive reduction in tumor TH, and the tumor was under control.

Overall, 10 mL of peripheral blood was collected from all the patients in Norgen cfDNA blood collection tubes (Cat. 63950) at each phase. The time it took for the blood collection and cfDNA isolation was 2–3 days. According to studies, the half-life of cfDNA in circulation is between 16 min and 2.5 h [37, 38]. After centrifugation (400 g for 10 min, 12,000 g for 10 min, 20,000 g for 10 min), at least 4 mL of plasma was collected for the next step. The supernatants were collected and kept at -80°C until DNA extraction. cfDNA was extracted using the HiPure Circulating DNA Midi Kit C (Magen Biotech, China). Then, cfDNA concentration was measured using a Qubit 2.0 Fluorometer (Thermo Fisher, Paisley, UK) with dsDNA High Sensitivity Assay Kit.

After ensuring the proper concentration, the preparation was performed using the AmpliSeq libraries. Sequencing library preparation is provided by Ion AmpliSeq Cancer Hotspot Panel v2, which contains a set of primers designed to probe hotspots in genes commonly mutated in human cancer. This panel targets 50 oncogenes and TSGs to amplify 207 amplicons, covering nearly 2,800 COSMIC mutations (<https://tools.lifetechnologies.com/content/sfs/brochures/Ion-AmpliSeq-Cancer-Hotspot-Panel-Flyer.pdf>, Thermo Fisher). Out of the 50 genes in this panel, 29 genes are related to UM (online suppl. Table S1). The manufacturer's protocol was used for the preparation of Ion AmpliSeq DNA libraries (Life Technologies, Carlsbad, CA, USA). Multiplex PCR was done with the Ion AmpliSeq Library Kit 2.0 using 3 ng DNA and the panel single primer pool. Using the FFPE extraction kit (QIAamp DNA FFPE Tissue Kit, QIAGEN), we extracted genomic DNA from a paraffin-embedded retinal tissue section of patient 1. Then, to look for copy number variations, we conducted the MLPA-UM kit (P027-C2, MRC-Holland, Amsterdam, The Netherlands), according to the manufacturer's protocol.

The data from the Ion Proton Sequencer (Thermo Fisher Scientific – US) were first run using the Ion Torrent platform-specific pipeline software, followed by reprocessing to read the sequences, cut the adapter sequences, filter, and retransfer weak signals. The sequences were aligned to the *Homo sapiens* hg19 reference genome and stored as BAM files.

Initial variant calling from Ion AmpliSeq sequencing data was performed using Torrent Suite version 5.12 software with the “variant caller” plugin program using key parameters: minimum allele frequency = 0.1 for variant calling of single nucleotide polymorphisms and 5% for small insertions and deletions (indels), quality control >99.99%, mean read depth >1,500× with coverage >100 reads, and p value <0.01. Sequencing data provided a Variant Call Format with a list of variants for the final clinical report. The open-source FRANKLIN database (<https://franklin.genoox.com>) takes the Variant Call Format file and the patient's phenotypes. All variants were analyzed through the Franklin Genoox platform to obtain proposed classifications using the American College of Medical Genetics and Genomics (ACMG) and the Association for Molecular Pathology (AMP) guidelines for interpretation and evidence associated with each variant.

According to the ACMG guidelines, the variants are classified into five main categories, including pathogenic, likely pathogenic, uncertain significance, likely benign, and benign. We also classified the sequence variants in somatic conditions into four categories based on their clinical effect according to AMP guidelines: tier I, variants with strong clinical significance; tier II, variants with potential clinical significance; tier III, variants with unknown clinical significance; and tier IV, variants that are benign or probably benign.

Furthermore, from a wide range of filter parameters, including variant annotation (chromosome, effect), variant calling (read coverage, genotype quality), internal frequency (minor allele frequency), and filters using external population datasets (gnomAD, 1,000 Genomes), variant type prediction is done via dbSNV, Splice AI, VARITY, MetaLR, and Revel; and clinical interpretation is conducted through ClinVar, ClinGen, GeneReviews, and OMIM databases.

In this study, we chose pathogenic, likely pathogenic, and uncertain significance for germinal classification and tier I, tier II for somatic classification. Next, genetic changes potentially effective in malignancy (potentially actionable findings) were identified and prioritized based on the level of pathogenicity. Finally, variants that were placed in polynucleotide sites and missing data (no calls) were excluded during the analysis.

To contextualize our findings and strengthen the analysis, we also conducted a systematic review of published studies on ctDNA in UM, identifying an additional 32 cases [39–42]. The electronic databases of PubMed, Scopus, Embase, and Google Scholar were utilized to find eligible papers published up to June 2024. Thus, together with the 3 patients from our study, a total of 35 UM patients were included in this study.

The serial analysis of ctDNA levels of patients with UM after applying filters (somatic classification, germinal classification, region, effect, and variant allele frequency above 0.1%) [43] revealed 10 important variants out of 50 genes in the panel. According to the KEGG pathway database and GeneCards, the important genes obtained were divided into two groups: oncogenic (70% [7/10]) and TSGs (30% [3/10]), and all of them are located in the MAPK and PI3K/AKT/mTOR signaling pathways (Table 1). Subsequently, these important genes were analyzed in each patient at each phase.

In patient 1, during the time of recurrence, the baseline variants related to UM recurrence were *GNAS*: c.602G>A, *IDH1*: c.394C>T, *GNAQ*: c.626A>G, and *FGFR2*: c.1144T>C (Fig. 1). In the 6th month, the VAF for all variants was undetectable. In the 12th month, the VAF in *GNAS* increased and then decreased in the last phase (VAF = 0.05, allele depth = 2,000). In the last phase, the VAF in the other variants (*IDH1*, *GNAQ*, and *FGFR2*) increased again. The reason for the decrease in *GNAS* in the third phase may be due to the elimination of the subclones containing this variant by the dominant subclones due to positive selection [44].

VAF in other variants related to UM, including *STK11*: c.996G>A, *EGFR*: c.2234A>G, *GNA11*: c.625C>A, *EZH2*: c.1937A>G, *ATM*: c.9139C>T, and *PTEN*: c.328C>T, increased in phases 3 (12th month) and 4 (24th month) (Fig. 1). Next, the tissue sample was subjected to MLPA for cytogenetic changes, and it was evident that the short arm of chromosome 8 (8p) was removed (online suppl. Fig. S6).

In patient 2, at the time of recurrence at the 12th month, variants related to UM recurrence were *GNAS*: c.601C>A, *IDH1*: c.395G>A, *GNAQ*: c.626A>C, *STK11*: c.109C>T, *EGFR*: c.2281G>A, *GNA11*: c.625C>A, and *EZH2*: c.1937A>G. After TTT (2 weeks later), except for *STK11* and *EGFR* variants, other variants decreased in the last phases (Fig. 2). In the 6th month, VAF for all variants was undetectable. In the 12th month, frequencies of *GNAS* increased, and then in the last phase it decreased (VAF = 0.05, depth = 2,000). In the last phase, VAF in other variants (*IDH1*, *GNAQ*, and *FGFR2*) re-increased.

VAF in other variants related to UM, including *ATM*: c.9139C>T, and *FGFR2*: c.1124A>G, increased in phases 3 and 4. However, *PTEN*: c.328C>T increased in phase 4 only (Fig. 2). By

Table 1. Composition of the gene panel and functional roles of selected genes associated with UM recurrence and metastasis

Gene	Functional role	Classification	Associated signaling pathway
GNAQ	G-protein signaling	Oncogene	MAPK, PI3K/AKT
GNA11	G-protein signaling	Oncogene	MAPK, PI3K/AKT
GNAS	G-protein signaling	Oncogene	cAMP, BAP1
IDH1	Metabolism regulator	Oncogene	Epigenetic regulation, BAP1
FGFR2	Receptor tyrosine kinase	Oncogene	MAPK, PI3K/AKT
EGFR	Receptor tyrosine kinase	Oncogene	MAPK, PI3K/AKT
EZH2	Epigenetic modifier	Oncogene	PRC2, chromatin remodeling
STK11	Kinase	Tumor suppressor	mTOR, AMPK
PTEN	Phosphatase	Tumor suppressor	PI3K/AKT
ATM	DNA damage response	Tumor suppressor	p53, DNA repair

GNAQ, guanine nucleotide-binding protein G(q) subunit alpha (Gαq); GNA11, guanine nucleotide-binding protein subunit alpha-11; GNAS, guanine nucleotide-binding alpha-stimulating subunit; IDH1, isocitrate dehydrogenase 1; FGFR2, fibroblast growth factor receptor 2; EGFR, epidermal growth factor receptor; EZH2, enhancer of zeste homolog 2; STK11, serine/threonine kinase 11; PTEN, phosphatase and tensin homolog; ATM, ataxia-telangiectasia mutated; MAPK, mitogen-activated protein kinase; PI3K/AKT, phosphoinositide 3-kinase/protein kinase B; cAMP, cyclic adenosine monophosphate; BAP1, BRCA1-associated protein 1; PRC2, polycomb repressive complex 2; mTOR, mammalian target of rapamycin; AMPK, AMP-activated protein kinase; p53, tumor protein p53.

applying TTT in phase 3 after tumor recurrence, we observed a decrease in ctDNA level in the last phase (*GNAS*, *IDH1*, *GNAQ/11*, and *EZH2*).

In patient 3 (a nonrecurrent patient), the pathogenic variants related to UM invasion (*GNAS*, *GNA11*, *EZH2*, *IDH1*, and *ATM*) were negative and the VAF in other important genes (*STK11*: c.109C>T, *EGFR*: c.2356G>A, *FGFR2*: c.1124A>G, and *PTEN*: c.992A>G) decreased from baseline to the last phase (Fig. 3). Finally, we found that only the co-occurrence of three genes, including *GNAQ*, *GNAS*, and *IDH1*, can be considered as important genes involved in UM recurrence (Table 2).

Next, using the BioGRID 4.4 database, the interaction between proteins was drawn online. The *GNAS* protein interacted with the *GNAQ/11* and *BAP1* proteins (online suppl. Fig. S7), and *IDH1* interacted with the *BAP1* protein (online suppl. Fig. S8).

By exploring recent studies [39–42], we encountered a series of genes that were identified with different ctDNA levels at the time of diagnosis, recurrence, and metastasis of UM (Table 3). Among the 35 patients, the most important genes involved at the time of recurrence were *GNAS* (p.R201H/C), *GNAQ/11* (Q209P/R), *IDH1* (R132H/C), *FGFR2* (C382A), *STK11* (Gln37*), and *EGFR* (D761Y). At the time of metastasis, the genes included *GNAS* (p.R201H), *GNAQ/11* (Q209), *FOXO1* (p.S22W), *SF3B1* (R625C), and *EIF1AX* (R13H) (Table 3). All of these known variants at the time of recurrence and metastasis are directly or indirectly related to the *BAP1* signaling pathway [8, 45].

Discussion

Our study detected ctDNA alterations before and after treatment of TTT, BRT, and enucleation for two recurrent and one nonrecurrent UM in four phases. VAF was checked using Ion Proton in the cfDNA, and the proportion of mutant DNA to the total number

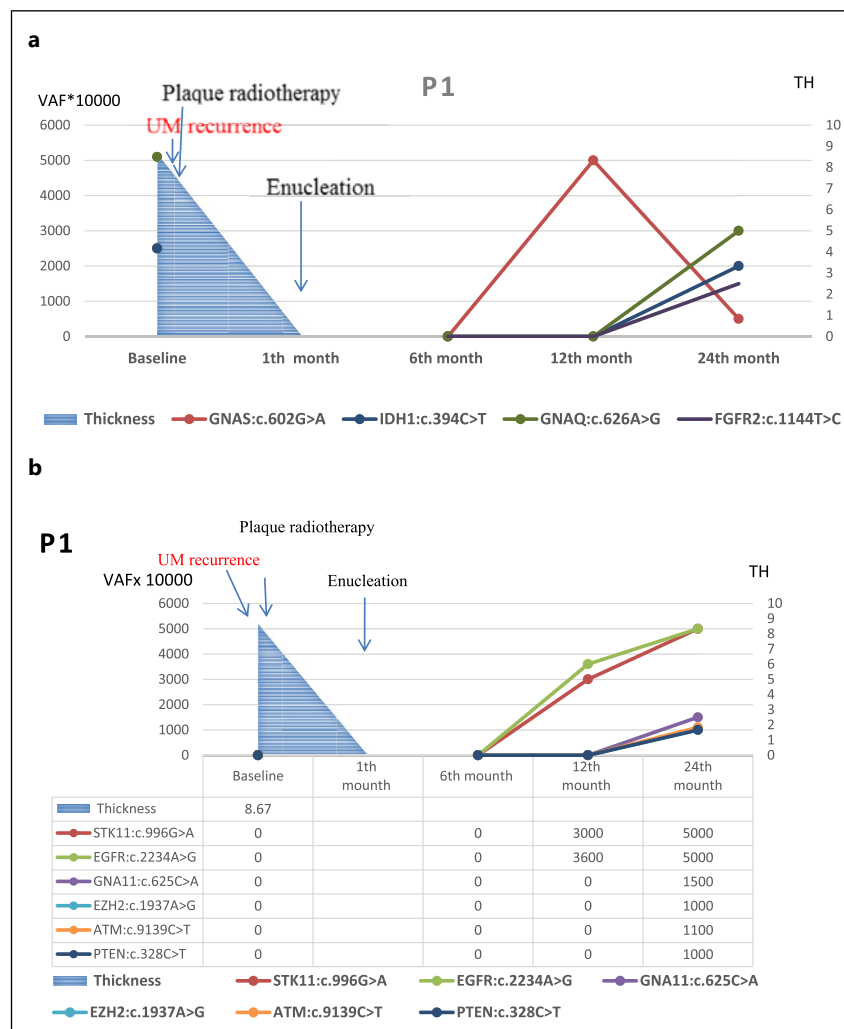


Fig. 1. a Line graph showing tumor thickness (TH) and variant allele frequency (VAF) changes over time for patient P1. Mutations include GNAS:c.602G>A, IDH1:c.394C>T, GNAQ:c.626A>G, and FGFR2:c.1144T>C. TH decreases from baseline (8.67) to 24 months (value not shown). VAF values for mutations decline over time. Plaque radiotherapy, UM recurrence, and enucleation are indicated. **b** Mutations include STK11:c.996G>A, EGFR:c.2234A>G, GNA11:c.625C>A, EZH2:c.1937A>G, ATM:c.9139C>T, and PTEN:c.328C>T. TH remains stable around 8.67. VAF values increase for most mutations by 24 months. UM recurrence, plaque radiotherapy, and enucleation are indicated.

(mutated + wild-type sequences) represents a percentage alteration in the plasma of the patients.

Interestingly, after analyzing the 50-gene panel, we identified 10 important genes involved in UM recurrence, including *GNAQ*, *GNA11*, *GNAS*, *EGFR*, *FGFR2*, *IDH1*, and *EZH2* as oncogenic genes and *STK11*, *PTEN*, and *ATM* as suppressor genes, that are part of the UM pathway. Several studies have shown that these genes are involved in UM tumorigenesis [43, 46–49]. Only two studies to date have shown that UM tumors harbor *IDH1* mutations with variants V178I [48] and R132C [49].

We identified that, similar to other studies, after BRT and enucleation, ctDNA has a positive effect on variant load [37, 38]. Our findings suggest that enucleation may transiently influence ctDNA levels due to tumor cell disruption; however, in our enucleated patient,

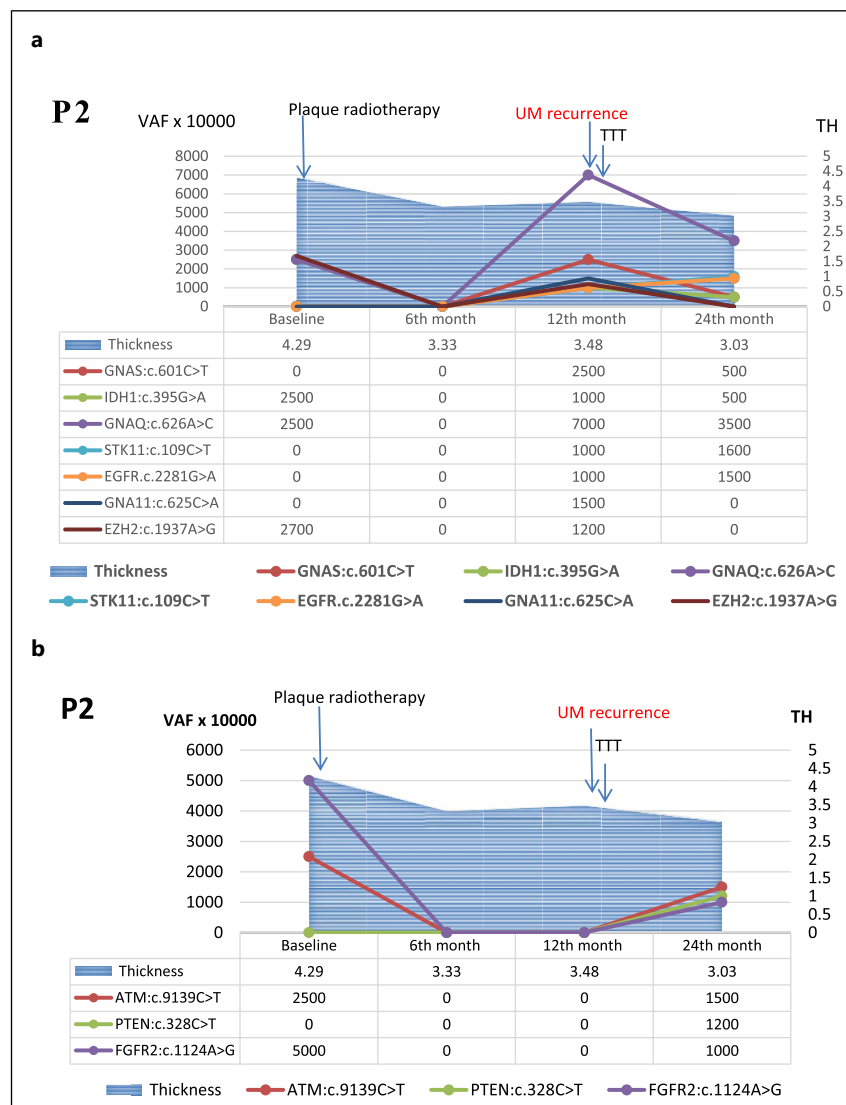


Fig. 2. a Line graph showing changes in tumor thickness (TH) and variant allele frequency (VAF) over time for patient P2. TH decreases slightly from baseline (4.29) to 24 months (3.03). Mutations include GNAS:c.601C>T, IDH1:c.395G>A, GNAQ:c.626A>C, STK11:c.109C>T, EGFR:c.2281G>A, GNA11:c.625C>A, and EZH2:c.1937A>G. Some mutations decrease or disappear over time, while GNAQ:c.626A>C increases. Plaque radiotherapy and UM recurrence are indicated. **b** TH remains stable around 3–4. Mutations include ATM:c.9139C>T, PTEN:c.328C>T, and FGFR2:c.1124A>G. ATM mutation appears at 24 months, while FGFR2 is present at baseline and reappears at 24 months. Plaque radiotherapy and UM recurrence are indicated.

variant load decreased postoperatively. Therefore, enucleation does not inherently increase ctDNA positivity long term. Instead, persistent or rising ctDNA levels post-enucleation may suggest residual or systemic disease rather than a direct effect of the surgical procedure itself. Additionally, our study revealed that after TTT, the load of some variants was also decreased.

By comparing ctDNA variants in two recurrent and one nonrecurrent UM cases, we observed that the co-occurrence of GNAQ (Q209P/R), GNAS (R201H/C), and IDH1 (R132H/C) was associated with recurrence. These genes are part of the BAP1 pathway, which has

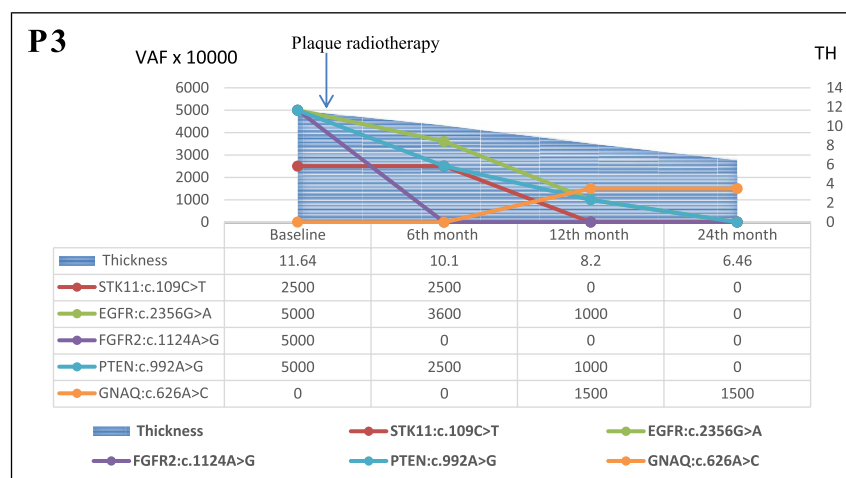


Fig. 3. Line graph showing tumor thickness (TH) and variant allele frequency (VAF) changes over time for nonrecurrent patient P3. TH decreases from 11.64 at baseline to 6.46 at 24 months. Mutations include STK11:c.109C>T, EGFR:c.2356G>A, FGFR2:c.1124A>G, PTEN:c.992A>G, and GNAQ:c.626A>C. Most mutations decrease or disappear by 12–24 months, while GNAQ appears later. Plaque radiotherapy is indicated.

been previously linked to tumor progression in UM. Jiang et al. [50] have proved that somatic co-mutation is one of the mechanisms of tumorigenesis. However, one study showed that the co-occurrence of BAP1 and SF3B1 mutations has an inhibitory effect on UM invasion [45].

In the patient with no recurrent lesion, the pathogenic variants *GNAS*, *GNA11*, *EZH2*, *IDH1*, and *ATM* related to UM invasion were negative. Additionally, an assessment of important genes in this patient with no recurrent lesion showed a descending trend of VAF from baseline to the 24th month.

Our results showed that there is a probability of UM recurrence with the minimum co-occurrence VAF in *GNAQ*, *GNAS*, and *IDH1* of 0.51%, 0.25%, and 0.1%, respectively. If ctDNA levels show a rising trend (especially in high-risk variants such as *GNAQ*, *GNAS*, and *IDH1*) but imaging and clinical examinations remain negative, we propose a more intensive surveillance strategy. This may include shortening the imaging interval (e.g., MRI and CT every 2–3 months), reassessment of tumor markers, and potential use of adjunct therapies such as TTT. While invasive intervention may not be immediately warranted, molecular positivity could signal early tumor activity that precedes radiologic detectability.

Several studies on cutaneous melanoma have shown that ctDNA, in addition to being correlated with tumor size and accurately indicating tumor mutations, can also detect recurrence [51, 52] and treatment failure prior to radiological imaging [53–55]. Additionally, ctDNA detection can be helpful in identifying stage III cutaneous melanoma patients at risk of recurrence [56, 57]. Similar to systemic (cutaneous) melanomas, ctDNA in UM reflects tumor burden, mutation status, and recurrence. However, UM exhibits distinct molecular features (e.g., *GNAQ*/*GNA11* mutations instead of *BRAF*/*NRAS*) and has a different metastatic pattern (predominantly hepatic). The utility of ctDNA in both cancers includes early recurrence detection and monitoring treatment response, but UM requires tailored panels due to its unique mutation spectrum. Notably, the ctDNA VAF thresholds associated with recurrence in UM appear to be lower than in cutaneous melanoma.

Francis et al. [39] reported that when the cancer recurrence occurred, there was no ctDNA present, and MRI showed no signs of metastatic disease. After 3 months, a follow-up ctDNA test detected *GNAQ* p.Q209P (VAF = 1.49%) and *EIF1AX* p.R13H (VAF = 1.14%) mutations and the development of a new liver metastasis.

Table 2. Important genes involved in UM recurrence

Genes	VAF	
	patient 1 (UM recurrence: baseline)	patient 2 (UM recurrence: 12 months)
GNAQ (Q209P/R)	0.51%	0.70%
GNAS (R201H/C)	0.25%	0.25%
IDH1(R132H/C)	0.25%	0.1%

Francis et al. [40] evaluated the presence of ctDNA in the plasma of UM patients and found that 2 patients had a *GNAS* mutation. However, a patient with metastatic disease had an initial ctDNA VAF of 0.29% and 0.72% around the time of metastases. In contrast, our study showed that the ctDNA VAF in *GNAS* was 0.25% at the time of recurrence. This may indicate that the desired cutoff for metastasis development has not yet been achieved.

Francis et al. [41] described a man with metastatic UM who had disease progression after hepatic embolization and nivolumab/ipilimumab. Plasma ctDNA GNA11 p.Q209L (VAF 0.15%) and SF3B1 p.R625C (VAF 0.38%) were measurable and increased after progression on vorinostat. Additional nivolumab/ipilimumab led to a radiographic response, with undetectable ctDNA at 27 months of follow-up.

Le Guin et al. [42] reported that recurrence/metastatic patients prior to the clinical diagnosis of disease progression had a positive ctDNA biomarker (GNAQ or GNA11), ranging between 2 and 10 months (mean 5.7 months). The minimum VAF for recurrence/metastatic patients was reported as 0.13%. However, they did not evaluate the VAF trend from the baseline to the 24th month in different phases.

Two known risk factors for tumor recurrence are tumor size (largest diameter and tumor TH) and tumor location (commonly the posterior position) [16, 58, 59]. In our study, 2 recurrent patients had an increase in tumor size at the time of recurrence and the location of the tumor was at the posterior fundus but in a patient with no recurrent UM located in the anterior position.

In addition, monosomy 8p was observed in a case that received BRT twice and was enucleated after recurrence. Loss of the 8p12-22 region is associated with rapid metastasis in primary UMs with aggressive morphology due to the deletion of multiple genes such as LZTS1 [60, 61].

The main limitations of this study are the relatively short follow-up time, the small number of investigated patients due to budget limitations, and the lack of normal tissue in patients as a control. Nevertheless, our experience confirms that the presence of ctDNA levels with an increasing trend may be a harbinger of recurrent disease. Early detection of patients with recurrent lesions and subsequent management may increase the anatomical success rate following BRT.

The CARE Checklist has been completed by the authors for this case report, attached as online supplementary material. In conclusion, this study focuses on the evaluation of ctDNA levels in UM patients at the time of diagnosis, recurrence, and metastasis using the currently available literature. The co-occurrence of GNAQ, GNAS, and IDH1 variants might be an important biomarker for UM recurrence after treatment. The most important genes at the time of recurrence include *GNAS*, *GNAQ/11*, *IDH1*, *FGFR2*, *STK11*, and *EGFR*, while at the time of metastasis, they include *GNAS*, *GNAQ/11*, *FOXO1*, *SF3B1*, and *EIF1AX*, which are directly or indirectly related to the BAP1 signaling pathway. The increase in the levels of these variants might be related to the development and progression of recurrent disease.

Table 3. ctDNA detection studies in UM

Study	Methods	Patients	Type of mutations	VAf at diagnosis, %	VAf at recurrence, %	VAf at metastasis, %
Francis et al. [39] (2021)	MSK-ACCESS assay	1	GNAQ exon5 p.Q209P; EIF1AX p.R13H	N/A	ND	1.49; 1.14
Francis et al. [40] (2022)	MSK-ACCESS assay (using a 129-oncogene panel using next-generation deep sequencing and hybridization capture)	1	TP53 exon5 p.A161V	1.29	No rec	No mets
		2	NF1 exon2 p.N1156D	0.32	No rec	No mets
		3	SF3B1 exon15 p.K700E	0.59	No rec	No mets
		4	SF3B1 exon14 p.K666R	1.42	No rec	No mets
		5	GNAQ exon5 p.Q209L; NF1 exon27p.G1168D	0.37; 0.25	No rec	No mets
		6	GNAQ exon5 p.Q209L; KEAP1 exon3 p.C395Y	0.32; 0.28	No rec	No mets
		7	GNAS exon8 p.R201H	0.29	No rec	> 0.53
		8	SF3B1 exon14 p.R625C	0.16	No rec	No mets
		9	FOXO1 exon1 p.S22w	0.25	No rec	No mets
		10	CDKN2A exon2 p.D108N	0.13	No rec	No mets
		11	GNA11 exon5 p.Q209L; SF3B1 exon14 p.R625H	1.22; 1.27	No rec	No mets
		12	GNAQ exon5 p.Q209L; EIF1AX exon2 p.R13P	0.88; 0.56	No rec	No mets
		13	FOXO1 exon1 p.S22w	1.26	No rec	No mets
		14	FOXO1 exon1 p.S22w	N/A	No rec	>0.91
Francis et al. [41] (2024)	MSK-ACCESS assay	1	GNA11 exon5 p.Q209L SF3B1 exon 14 p.R625C	ND	ND	0.15; 0.38
Le Guin et al. [42] (2021)	Deep amplicon sequencing of GNAQ/GNA11 mutations	17	GNAQ or GNA11 alleles	>0.1	>0.1	>0.1
This study	Ion Proton Sequencer using the Ion AmpliSeq Cancer Hotspot Panel v2 (50 oncogenes and TSGs)	1	GNAS exon8 p.Arg201His	N/A	0.25	No mets
			IDH1 exon1 p.Arg132Cys	N/A	0.25	No mets
			GNAQ exon5 p.Gln209Arg	N/A	0.51	No mets
			FGFR2 exon9 p.Cys382Arg	N/A	0.25	No mets
		2	GNAS exon8 p.Arg201Cys	N	0.25	No mets
			IDH1 exon1 p.Arg132His	0.25	0.1	No mets

(Continued on following page)

Table 3 (continued)

Study	Methods	Patients	Type of mutations	VAF at diagnosis, %	VAF at recurrence, %	VAF at metastasis, %
			GNAQ exon5 p.Gln209Pro	0.25	0.70	No mets
			STK11 exon1 p.Gln37*	N	0.10	No mets
			EGFR exon19 p.Asp761Tyr	N	0.10	No mets
			GNA11 exon5 p.Gln209Lys	N	0.15	No mets
			EZH2 exon16 p.Tyr646Cys	0.27	0.12	No mets
			ATM exon = 63 p.Arg3047*	0.25	ND	No mets
			FGFR2 exon9 p.Tyr375Cys	0.50	ND	No mets
		3	STK11 exon1 p.Gln37*	0.25	No rec	No mets
			EGFR exon20 p.Val786Met	0.50	No rec	No mets
			FGFR2 exon9 p.Cys382Arg	0.50	No rec	No mets
			PTEN exon8 p.Asp331Gly	0.50	No rec	No mets

To prepare the best panel and optimal VAF cutoff for the early diagnosis of UM recurrence, further studies in a larger cohort size with longer follow-up are required to confirm the implications of related genes in UM tumorigenesis.

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Statement of Ethics

This study was performed in accordance with the Declaration of Helsinki. This human study was approved by Iran university of Medical Sciences – approval: IR-IUMS.REC.1398.730. All adult participants provided written informed consent to participate in this study. Written informed consent was obtained from the individuals for publication of the details of their medical case and any accompanying images.

Conflict of Interest Statement

The authors declare that they have no conflict of interest.

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Author Contributions

F.A. wrote the initial draft of the manuscript, collected samples and performed genetic analysis, and interpreted genetic tests. S.T. suggested the main conceptual ideas and proofed outline and critically reviewed and revised the article for important intellectual content and contributed in the data. R.M. and G.K. revised the manuscript text. M.N. introduced and treated the patients and revised the final draft of the manuscript. All authors read and approved the final manuscript.

Data Availability Statements

All data generated or analyzed during this study are included in this article and its online supplementary material. Further inquiries can be directed to the corresponding author.

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